

COST ANALYSIS

- ◉ The concept of cost is a key concept in Economics.
- ◉ It refers to the amount of payment made to acquire any goods and services.
- ◉ In a simpler way, the concept of cost is a financial valuation of resources, materials, and utilities consumed to purchase goods and services.

TOTAL COSTS

- ⊙ Total Cost (TC) describes the total economic cost of production. It is composed of variable, and fixed, and opportunity costs.
- ⊙ Fixed costs
 - The accounting costs which **do not** change based on your level of output
 - Always determined to be fixed in the short term; if you could not change it on short notice it is fixed
 - **EXAMPLE** building costs, insurance, property taxes
- ⊙

◉ Variable costs

- The accounting costs that **do** change based on your output level
- Always determined in the short run (all factors are variable in the long run); if you could change it on short notice it is variable
- **EXAMPLE** number of widgets produced, number of low skilled employees, packaging costs

- ◉ **Formula,,,,,,,,,,,,,,,,,,,,,**
- ◉ Total Cost = Total Fixed Cost + Total Variable Cost
- ◉ $TC = TFC + TVC$

AVERAGE COSTS

◉ Definition

- Average cost is equal to total cost divided by the number of goods produced (the output quantity, Q).
- It is also equal to the sum of average variable costs (total variable cost divided by Q) plus average fixed costs (total fixed costs divided by Q).

◉ Here is a standard formulaic expression representing Average Costs:

- $AC = TC / Q$

MARGINAL COSTS

◉ Definition

- Marginal cost is the change in total costs that arises when the quantity produced changes by one unit.
- That is, it is the cost of producing one more unit of a good.
- Mathematically, the marginal cost (MC) function is expressed as the first derivative of the total costs (TC) function with respect to quantity (Q).

◉ Here is a standard formulaic expression representing Total Cost:

- $MC = dTC / dQ$

COSTS IN SHORT RUN

- ◉ **1. Average Fixed Cost (AFC)**
- ◉ The average fixed cost is the total fixed cost divided by the number of units produced. Hence, if TFC is the total fixed cost and Q is the number of units produced, then
- ◉ $AFC = TFC / Q$
- ◉ Therefore, AFC is the fixed cost per unit of output.
- ◉ **Example:**
- ◉ The TFC of a firm is Rs. 2,000. If the output is 100 units, the average fixed cost is,
- ◉ $AFC = TFC / Q = 2000 / 100 = \text{Rs. } 20$
- ◉ If the output is increased to 200 units, then
- ◉ $AFC = TFC / Q = 2000 / 200 = \text{Rs. } 10$
- ◉ Since TFC is constant, any increase in output decreases the AFC. Note that, while the AFC can become really small, it is never zero

2. AVERAGE VARIABLE COST (AVC)

- ◉ The second aspect of short-run average costs is an average variable cost. Average variable cost is the total variable cost divided by the number of units produced.
- ◉ Hence, if TVC is the total fixed cost and Q is the number of units produced, then
- ◉ $AVC = \{TVC\} / \{Q\}$
- ◉ Therefore, AVC is the variable cost per unit of output.
- ◉ Usually, the AVC falls as the output increases from zero to normal capacity output. Beyond the normal capacity, the AVC rises steeply due to the operation of diminishing returns.

3. AVERAGE TOTAL COST (ATC)

- ⦿ The average total cost is the sum of the average variable cost and the average fixed costs. That is,
- ⦿ $ATC = AFC + AVC$
- ⦿ In other words, it is the total cost divided by the number of units produced.
- ⦿ The diagram below shows the AFC, AVC, ATC, and Marginal Costs (MC) curves:

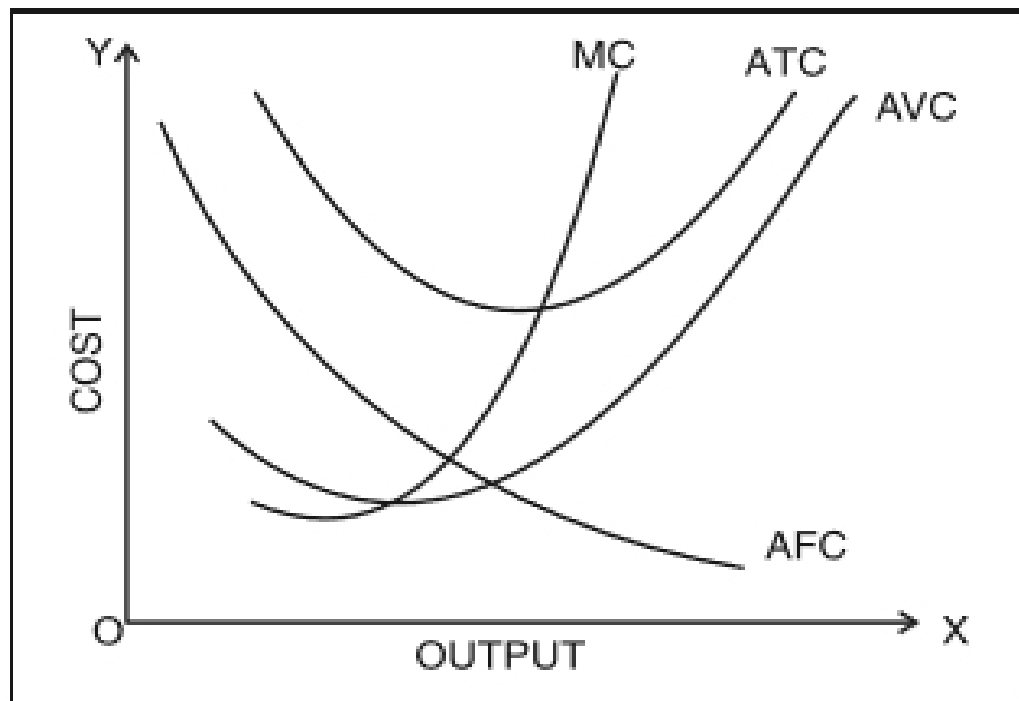


Fig. 1 : Short run Average and Marginal Cost Curves

Units of output	Total fixed cost	Total variable cost	Total Cost	Average fixed cost	Average variable cost	Average total cost	Marginal cost
0	150	0	150	-	-	-	-
6	150	50	200	25.0	8.33	33.33	$50/6 = 8.33$
16	150	100	250	9.38	6.25	15.63	$50/10 = 5.0$
29	150	150	300	5.17	5.17	10.34	$50/13 = 3.85$
44	150	200	350	3.41	4.55	7.95	$50/15 = 3.33$
55	150	250	400	2.73	4.55	7.27	$50/11 = 4.55$
60	150	300	450	2.50	5.0	7.50	$50/5 = 10.0$

- ◉ It is important to note that the behavior of the ATC curve depends upon that of the AVC and AFC curves. Observe that:
- ◉ In the beginning, both AVC and AFC curves fall. Hence, the ATC curve falls as well.
- ◉ Next, the AVC curve starts rising, but the AFC curve is still falling.
- ◉ Hence, the ATC curve continues to fall. This is because, during this phase, the fall in the AFC curve is greater than the rise in the AVC curve.
- ◉ As the output rises further, the AVC curve rises sharply.
- ◉ This offsets the fall in the AFC curve. Hence, the ATC curve falls initially and then rises.